

General

Information processing in the central nervous system occurs, among other things, by electrical activity of the nerve cells. This minimal, continuous electrical activity of the brain, which is produced by the brain itself, can be measured and recorded by using electrodes. The result of such a measurement is called an ElectroEncephaloGram, or EEG for short. EEG, alongside with measurement of eye signals and chin muscle activity with similar electrodes, is the 'gold standard' in sleep research. To allow for convenient sleep recordings at home, a wearable sleep EEG recorder can be applied at the Donders Institute and then taken home for the night.

Preparation at home

To make the EEG-measurement run more smoothly, you can run through the following steps:

- Wash and dry your hair beforehand;
- Do not use gel, hairspray, etc.;
- Do not use face cream or make-up;
- If needed, bring a comb or hair brush;



Preparation at the DCCN

About 15 single electrodes will be attached on your head, behind your ears, around your eyes and at your chin using water-soluble glue and/or small stickers. To obtain good signals it is important that the resistance of the skin is not too high. If necessary, the experimenter will make sure the resistance between your skin and the electrodes drops to the desired value by using some alcohol and conducting gel. The recorder will be attached with straps to your shoulder and chest above your shirt. In order to avoid changing clothes and thus needing to detach the recorder, please come already with a shirt/ pyjama that you will wear during the night.

The experiment

After this preparation, you will get instructed about what you have to do during the experiment and during the following night at home. You can sleep normally with the sleep recorder. It generally should not interfere with your sleep quality, although wearing the electrodes and recorder might feel a bit unaccustomed in the beginning. The recorder will run during the whole night, the measurement itself will not be noticeable for you. The experimenter will instruct you how to stop and detach the recorder in the morning. Electrodes can be cautiously detached from hair and skin by soaking the gel/glue with water for about a minute. Alternatively, you can also leave the electrodes attached and let the experimenter detach everything at the institute. In this case, if you want you can rinse out your hair, wash and dry it. For this purpose, shampoo and towels are available. For hygienic reasons, it is practical if you bring your own comb.

Additional information

The risk associated with participation can be considered as negligible risk and minimal burden. No invasive procedures are involved.

You can **NOT** participate in a EEG-experiment if one of the following applies:

- 1) You had head/brain surgery.
- 2) You suffer from epilepsy.
- 3) You suffer from claustrophobia.
- 4) You are pregnant or you think you are.
- 5) You are younger than 16 years of age.

If one of the above is applicable, please contact the researcher before the day of the experiment.